

HARRISBURG, PA, USA

WMO: 723990

Lat: 40.196N

Lon: 76.772W

Elev: 312

StdP: 14.53

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14711

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.4	14.8	-3.2	4.7	13.8	0.8	5.8	17.5	28.8	29.4	27.1	28.6	9.8	300	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	91.8	75.2	89.5	74.0	86.8	72.9	78.0	87.5	76.5	85.3	75.1	83.3	8.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	133.4	83.0	73.5	126.0	81.0	72.4	121.6	79.9	41.5	87.7	40.0	85.3	38.7	83.5	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.7	22.1	18.9	DB	5.0	96.2	7.1	3.1	-0.1	98.5	-4.3	100.3	-8.3	102.1	-13.4	104.3
			WB	3.7	80.8	6.7	1.9	-1.2	82.1	-5.1	83.2	-8.9	84.3	-13.8	85.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.4	30.4	33.1	41.8	53.4	63.3	72.5	77.0	75.1	68.1	56.1	44.8	35.8
	DBStd	17.90	9.69	8.95	9.04	8.11	7.50	6.09	4.90	4.79	6.51	7.72	7.84	8.07
	HDD50	2091	611	476	285	53	1	0	0	0	26	193	446	
	HDD65	5026	1073	892	720	357	125	12	0	1	41	295	606	904
	CDD50	3700	3	4	31	156	415	675	836	777	544	214	38	7
	CDD65	1160	0	0	1	10	74	237	372	313	135	18	0	0
	CDH74	9498	0	1	18	135	635	1921	3354	2433	889	111	1	0
	CDH80	3220	0	0	4	29	189	635	1294	797	252	20	0	0
	WSAvg	7.2	8.5	8.7	9.3	8.8	7.2	6.4	5.7	5.3	5.6	6.2	7.3	7.8
	PrecAvg	40.7	2.8	2.9	3.3	3.2	4.4	3.8	3.6	3.3	3.5	2.9	3.5	3.2
(15) Precipitation	PrecMax	59.3	8.0	5.9	6.1	8.0	9.7	18.5	9.7	7.8	15.0	9.9	7.2	7.6
	PrecMin	30.1	0.4	0.6	1.0	0.4	0.5	0.1	0.8	0.9	0.6	0.0	0.8	0.8
	PrecStd	6.9	1.7	1.4	1.3	1.6	2.2	3.3	2.0	1.6	2.9	1.8	1.6	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.6	64.5	75.4	83.8	89.8	92.9	97.3	93.5	90.6	82.4	70.6	64.6
		MCWB	57.6	53.7	59.8	65.1	71.6	75.4	75.7	76.0	73.7	69.7	59.5	58.2
	2%	DB	54.5	56.0	67.2	78.3	85.4	89.8	93.1	90.2	86.2	76.9	65.8	57.1
		MCWB	49.3	47.0	55.1	62.5	70.3	74.5	75.5	75.0	72.0	66.8	58.1	52.1
	5%	DB	47.5	50.2	61.4	72.5	81.2	86.5	90.0	87.2	82.4	72.5	61.4	51.6
		MCWB	42.8	43.2	51.5	59.1	67.6	72.7	74.4	73.5	70.2	64.5	54.1	47.5
	10%	DB	43.4	45.3	56.0	68.1	76.9	83.7	87.0	84.3	79.2	68.8	57.2	47.1
		MCWB	39.2	40.0	47.6	56.5	65.2	71.2	73.5	72.2	68.3	61.6	51.3	42.7
	0.4%	WB	59.0	57.5	61.7	68.2	73.9	78.4	80.1	80.1	77.1	73.1	63.9	59.3
		MCDB	61.5	62.5	74.8	77.5	85.1	88.3	91.0	89.5	85.9	78.2	67.5	63.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	50.3	48.2	57.2	64.6	71.7	76.5	78.1	77.7	74.6	69.6	59.8	53.7
		MCDB	53.5	54.7	64.8	73.7	82.2	85.9	88.4	85.9	81.9	74.9	64.1	56.7
	5%	WB	44.1	43.9	53.0	61.2	69.5	74.9	76.6	76.0	72.9	66.2	56.2	47.7
		MCDB	46.8	48.8	59.9	70.3	78.5	83.4	86.1	83.6	79.3	70.9	60.1	50.6
	10%	WB	39.6	40.6	48.6	58.1	67.4	73.2	75.2	74.5	71.0	62.7	52.4	43.6
		MCDB	42.8	45.3	55.5	66.2	74.7	81.0	83.8	81.4	76.8	67.4	56.2	46.8
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	15.5	17.3	19.5	18.6	17.6	17.5	17.0	17.6	17.7	16.1	13.3
		MCDBR	19.4	23.5	25.8	26.9	23.9	20.8	20.5	19.6	21.2	22.5	22.3	19.7
		MCWBR	15.7	16.6	16.0	14.2	11.6	9.5	8.1	8.4	10.0	13.0	14.9	15.9
		MCDBR	18.6	20.7	23.8	23.5	21.1	18.8	17.8	17.2	17.5	18.5	18.6	17.4
	5% WB	MCWBR	16.2	16.0	16.4	14.4	11.3	9.7	8.2	8.6	9.6	12.0	14.5	15.6
(40) Clear-Sky Solar Irradiance	taub		0.312	0.330	0.370	0.400	0.431	0.502	0.513	0.501	0.420	0.371	0.330	0.310
	taud		2.424	2.348	2.317	2.267	2.214	2.051	2.036	2.054	2.252	2.395	2.475	2.470
	Ebn at Noon		269	279	279	277	270	250	246	245	259	262	261	261
	Edh at Noon		26	32	37	42	45	53	53	51	39	30	24	23
(44) All-Sky Solar Radiation	RadAvg		578	861	1152	1493	1676	1856	1871	1637	1338	950	668	491
	RadStd		48	108	84	139	195	154	102	90	175	107	64	57

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	N/A	
(47) Regional (63 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+214	+117	

Nomenclature: See separate page